Question 3: First-Order Logic Multi-Clause Resolution Challenge

We have the following clauses:

1. $\exists z.R(z) \rightarrow \exists w.T(w)$
2. $\forall x \exists y.(\forall z.(P(x) \vee R(z)) \wedge \exists w.(T(w) \rightarrow Q(y)))$
3. $\forall w \exists y.(\forall x.(P(x) \rightarrow T(w)) \wedge \forall x.(Q(y) \rightarrow P(x)))$
4. $\forall x \exists z.(\neg R(z) \vee \neg Q(x))$
5. $\forall x.(U(x) \rightarrow (P(x) \vee \neg T(f(x))))$
6. $\exists x.(Q(x) \wedge \neg U(x))$
7. $\forall x \forall y.(\neg P(x) \vee Q(y) \vee S(x, y))$

Goal (G): $\forall x.P(x)$

Tasks:

- **(a)** Convert all into CNF. Show Skolemization and standardization.
- **(b)** Apply resolution to prove or refute G.
- **(c)** Identify (i) which clauses are redundant and (ii) which ones increase complexity the most. Provide justifications.
- **(d)** Suppose clause 6 is removed. Does this impact provability of G? Justify.

Part (a): Convert all into CNF. Show Skolemization and standardization.

> **Lecture Recap:**
> To convert any well-formed formula (wff) into Conjunctive Normal Form (CNF) clauses, we must strictly apply the following 9-step algorithm:
>
> 1. **Eliminate implication symbols:** Apply the equivalence $(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta)$ and $(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha))$.
> 2. **Reduce the scopes of negation symbols (¬):** Push negations inward using De Morgan's Laws $\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta)$ and $\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta)$, double negation $\neg(\neg\alpha) \equiv \alpha$, and quantifier equivalencies $\neg(\forall x)P(x) \equiv (\exists x)\neg P(x)$ and $\neg(\exists x)P(x) \equiv (\forall x)\neg P(x)$.
> 3. **Standardise variables:** Rename variables so that each quantifier binds a uniquely named variable.
> 4. **Eliminate existential quantifiers (∃) (Skolemise):** Replace existential variables with a **Skolem constant** if they do not depend on any universal quantifier (∀), or a **Skolem function** (e.g., $f(x)$) if they are inside the scope of a universal quantifier and thus depend on it.
> 5. **Move all universal quantifiers (∀) to the front.**
> 6. **Put the result in CNF:** Distribute OR (∨) over AND (∧) using the equivalency $(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma))$.
> 7. **Eliminate universal quantifiers (∀).**
> 8. **Eliminate conjunction symbols (∧):** Split the formula into separate, distinct clauses.
> 9. **Rename variables (Standardise variables apart):** Rename variables so no two final clauses share the exact same variable names.
>
> **Proof by Refutation:** To prove a goal using resolution, we must first negate the goal and add it to our knowledge base. If we can derive a contradiction (an empty clause, `NIL`), the original goal is proven.

Goal (G): $\forall x.P(x)$

- **Negate the Goal:** $\neg(\forall x.P(x))$
- **Step 2 (Reduce Negation):** Apply equivalence $\neg(\forall x)P(x) \equiv (\exists x)\neg P(x)$.
  Result: $\exists x.\neg P(x)$
- **Step 4 (Skolemise):** The variable x is not dependent on any universal quantifier. We replace x with a **Skolem constant** C.
  Result: $\neg P(C)$
- **Steps 5-9:**
  Clause G.1: $\neg P(C)$

Clause 1: $\exists z.R(z) \Rightarrow \exists w.T(w)$

- **Step 1 (Eliminate ⇒):** Apply $(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta)$.
  Result: $\neg(\exists z.R(z)) \vee \exists w.T(w)$

- **Step 2 (Reduce $\neg$):** Apply $\neg(\exists z)P(z) \equiv (\forall z)\neg P(z)$.
  Result: $(\forall z.\neg R(z)) \vee \exists w.T(w)$
- **Step 3 (Standardise):** Variables z and w are already unique.
- **Step 4 (Skolemise):** The existential quantifier $\exists w$ is strictly outside the scope of $\forall z$. It depends on nothing. We replace w with a **Skolem constant** A.
  Result: $(\forall z.\neg R(z)) \vee T(A)$
- **Step 5 (Move $\forall$ outward):**
  Result: $\forall z.(\neg R(z) \vee T(A))$
- **Step 6 & 7 (CNF & Drop $\forall$):**
  Result: $\neg R(z) \vee T(A)$
- **Step 9 (Standardise apart):** Rename z to z_1.
  Clause 1.1: $\neg R(z_1) \vee T(A)$

Clause 2: $\forall x \exists y.(\forall z.(P(x) \vee R(z)) \wedge \exists w.(T(w) \rightarrow Q(y)))$

- **Step 1 (Eliminate $\rightarrow$):** Apply $(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta)$ to the inner implication.
  Result: $\forall x \exists y.(\forall z.(P(x) \vee R(z)) \wedge \exists w.(\neg T(w) \vee Q(y)))$
- **Step 2 (Reduce $\neg$):** No negations to push inward.
- **Step 3 (Standardise):** Variables x, y, z, w are already uniquely bound by their respective quantifiers.
- **Step 4 (Skolemise):**
  - The existential variable y is strictly inside the scope of the universal variable x. We replace y with a **Skolem function** $f_1(x)$.
  - The existential variable w is also inside the scope of x, but crucially, it is *outside* the scope of $\forall z$. Therefore, w depends *only* on x, not z. We replace w with a **Skolem function** $f_2(x)$.
    Result: $\forall x.(\forall z.(P(x) \vee R(z)) \wedge (\neg T(f_2(x)) \vee Q(f_1(x))))$
- **Step 5 (Move $\forall$ outward):**
  Result: $\forall x \forall z.((P(x) \vee R(z)) \wedge (\neg T(f_2(x)) \vee Q(f_1(x))))$
- **Step 6 (CNF):** The formula is already a conjunction of disjunctions.
- **Step 7 (Drop $\forall$):**
  Result: $(P(x) \vee R(z)) \wedge (\neg T(f_2(x)) \vee Q(f_1(x)))$
- **Step 8 (Split into clauses):**
  Clause 2a: $P(x) \vee R(z)$
  Clause 2b: $\neg T(f_2(x)) \vee Q(f_1(x))$
- **Step 9 (Standardise apart):**
  Clause 2.1: $P(x_2) \vee R(z_2)$
  Clause 2.2: $\neg T(f_2(x_3)) \vee Q(f_1(x_3))$

Clause 3: $\forall w \exists y.(\forall x.(P(x) \rightarrow T(w)) \wedge \forall x.(Q(y) \rightarrow P(x)))$

- **Step 1 (Eliminate $\rightarrow$):**
  Result: $\forall w \exists y.(\forall x.(\neg P(x) \vee T(w)) \wedge \forall x.(\neg Q(y) \vee P(x)))$
- **Step 3 (Standardise):** The variable x is quantified twice in parallel scopes. To safely move the universal quantifiers outward in Step 5 without scope collision, we must rename the second x to a new variable, v.
  Result: $\forall w \exists y.(\forall x.(\neg P(x) \vee T(w)) \wedge \forall v.(\neg Q(y) \vee P(v)))$
- **Step 4 (Skolemise):** y depends on $\forall w$. Replace y with a **Skolem function** $f_3(w)$.
  Result: $\forall w.(\forall x.(\neg P(x) \vee T(w)) \wedge \forall v.(\neg Q(f_3(w)) \vee P(v)))$
- **Step 5 (Move $\forall$ outward):**
  Result: $\forall w \forall x \forall v.((\neg P(x) \vee T(w)) \wedge (\neg Q(f_3(w)) \vee P(v)))$
- **Steps 6, 7 & 8 (CNF, Drop $\forall$ & Split):**
  Clause 3a: $\neg P(x) \vee T(w)$
  Clause 3b: $\neg Q(f_3(w)) \vee P(v)$
- **Step 9 (Standardise apart):**
  Clause 3.1: $\neg P(x_4) \vee T(w_4)$
  Clause 3.2: $\neg Q(f_3(w_5)) \vee P(v_5)$

Clause 4: $\forall x \exists z.(\neg R(z) \vee \neg Q(x))$

- **Step 4 (Skolemise):** z depends on x. We use a **Skolem function** $f_4(x)$.
  Result: $\forall x.(\neg R(f_4(x)) \vee \neg Q(x))$
- **Steps 7-9 (Drop $\forall$, Split, Standardise apart):**
  Clause 4.1: $\neg R(f_4(x_6)) \vee \neg Q(x_6)$

Clause 5: $\forall x.(U(x) \rightarrow (P(x) \vee \neg T(f(x))))$

- **Step 1 (Eliminate $\rightarrow$):**
  Result: $\forall x.(\neg U(x) \vee P(x) \vee \neg T(f(x)))$
- **Steps 7-9 (Drop $\forall$, Split, Standardise apart):**
  Clause 5.1: $\neg U(x_7) \vee P(x_7) \vee \neg T(f(x_7))$

Clause 6: $\exists x.(Q(x) \wedge \neg U(x))$

- **Step 4 (Skolemise):** $\exists x$ is not inside any universal scope. Replace x with a **Skolem constant** B.
  Result: $Q(B) \wedge \neg U(B)$
- **Steps 8 & 9 (Split & Standardise):** No variables remain to Standardise.
  Clause 6.1: $Q(B)$
  Clause 6.2: $\neg U(B)$

Clause 7: $\forall x \forall y.(\neg P(x) \vee Q(y) \vee S(x, y))$

- **Steps 7-9 (Drop $\forall$, Split, Standardise apart):**
  Clause 7.1: $\neg P(x_8) \vee Q(y_8) \vee S(x_8, y_8)$

Part (b): Apply resolution to prove or refute G

> **Lecture Recap:**
>
> The **Resolution Inference Rule** combines two clauses by eliminating a pair of complementary literals (L and $\neg L$) using a Most General Unifier (MGU) substitution θ. The resolvent is formed by joining the remaining literals. We aim to derive an empty clause (NIL), meaning a contradiction is found.

We can achieve the proof rapidly using just four clauses from our knowledge base:

1. **Resolve Clause 6.1 with Clause 4.1:**
   - **Parent Clauses:** $Q(B)$ and $\neg R(f_4(x_6)) \vee \neg Q(x_6)$
   - **Complementary Literals:** Q and $\neg Q$
   - **Unification (MGU):** Substitute x_6 with constant B. $\theta_1 = \{x_6/B\}$.
   - **Resolvent 8:** $\neg R(f_4(B))$
2. **Resolve Resolvent 8 with Clause 2.1:**
   - **Parent Clauses:** $\neg R(f_4(B))$ and $P(x_2) \vee R(z_2)$
   - **Complementary Literals:** $\neg R$ and R
   - **Unification (MGU):** Substitute variable z_2 with the function $f_4(B)$. $\theta_2 = \{z_2/f_4(B)\}$.
   - **Resolvent 9:** $P(x_2)$
3. **Resolve Resolvent 9 with Negated Goal (G.1):**
   - **Parent Clauses:** $P(x_2)$ and $\neg P(C)$
   - **Complementary Literals:** P and $\neg P$
   - **Unification (MGU):** Substitute x_2 with constant C. $\theta_3 = \{x_2/C\}$.
   - **Resolvent 10:** `NIL`

> 🎯 **Conclusion:** We successfully derived a contradiction (NIL) from the negated goal. Therefore, the original goal $\forall x.P(x)$ is **proven**.

Part (c): Clause Analysis

(i) Which clauses are redundant?

Clauses **1, 3, 5, 6.2, 7, and 2.2** are completely redundant.

Justification:
In logic, a literal is "pure" if it appears with only one polarity (either always positive or always negative) across all clauses in the knowledge base. Resolution requires unifying opposite polarities to eliminate them.
* Predicate $S(x, y)$ only appears *positively* (in Clause 7.1).
* Predicate U only appears *negatively* ($\neg U$ in Clauses 5.1 and 6.2).
Because these literals lack opposing polarities, any clause containing them can never be resolved away to generate an empty clause. Therefore, Clauses **5.1, 6.2, and 7.1** are strictly mathematically useless for a refutation proof. Furthermore, since we generated an elegant proof only utilizing Clauses 2.1, 4.1, 6.1, and G.1, Clauses **1, 3, and 2.2** are functionally redundant for this specific goal.

(ii) Which ones increase complexity the most?

Clauses **2, 3, 4, and 5** increase computational complexity the most.

Justification:
These clauses introduce **Skolem functions** ($f_1(x)$, $f_2(x)$, $f_3(w)$, $f_4(x)$, and $f(x)$). In automated theorem proving, functions exponentially inflate the *Herbrand Universe* (the set of all possible term substitutions, e.g., substituting x with $f(A)$, $f(f(A))$, etc.). This creates a massive, potentially infinite branching factor during the Unification algorithm, drastically slowing down the resolution search space. Clause 2 specifically increases complexity the most because its nested quantifiers force the formula to introduce two different Skolem functions *and* branch into two separate CNF clauses via distribution.

Part (d): Suppose clause 6 is removed. Does this impact provability of G?

Yes, removing Clause 6 makes G strictly unprovable.

Justification:
Resolution-refutation requires grounding out logical chains with definitive, unconditional facts. Clause 6 provides the *only* standalone positive fact in the entire knowledge base: $Q(B)$ (Clause 6.1).

- If we remove Clause 6, we strip the knowledge base of its only unconditioned positive Q. To refute the negated goal $\neg P(C)$, we must resolve it against a clause that provides a positive P, such as Clause 2.1 ($P(x_2) \vee R(z_2)$). However, this resolution leaves a residual "debt" of positive literals (like $R(z_2)$) that must be canceled.
- If you trace the dependencies, canceling $R(z_2)$ using Clause 4.1 ($\neg R \vee \neg Q$) requires a positive Q to cancel the resulting $\neg Q$. Without Clause 6.1, the only other sources of a positive Q are Clause 7.1 (which is a pure literal trap due to S) and Clause 2.2 ($\neg T \vee Q$). Using Clause 2.2 introduces $\neg T$, which demands a positive T, introducing new negative literals ($\neg R$ or $\neg P$), throwing the inference engine into an infinite loop.
- Furthermore, attempting to resolve the remaining $\neg Q(f_3(w_5))$ (from 3.2) against $Q(f_1(x_3))$ (from 2.2) requires unifying distinct Skolem functions (f_1 and f_3), which mathematically fails because different functions cannot be unified. Without Clause 6.1, the algorithm is permanently starved of the positive facts required to reach `NIL`.